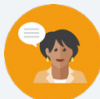


10.7 Comparing Data Sets and Graphs

Lesson Introduction

 Benchmark	MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		 Learning Target	I can compare and interpret a numerical representation of data of one population and a graphical representation of another population.
 Benchmark Coherence	Building On			Working Toward
	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.			MA.912.DP.2.1 For two or more sets of numerical univariate data, calculate and compare the appropriate measures of center and measures of variability, accounting for possible effects of outliers. Interpret any notable features of the shape of the data distribution.
 Essential Questions	- Is it easier to compare data presented graphically or in numerical data sets? Why? - How can you compare data sets without finding all the measures of center and variation? - How can you use measures of center and variation to draw conclusions about different data sets?			
 Lesson Narrative	In this lesson, students compare data presented numerically and graphically. Students begin by summarizing data presented numerically and in a box plot, considering the benefits and drawbacks of presenting data numerically and graphically. Students then participate in a dice-rolling activity to collect their own data and compare it to a partner's. The lesson culminates with students summarizing two data sets to compare them and draw conclusions.			
 Component Summary	10.7.1 Warm-Up (~5 min.)	Students choose between two deals (<i>SE p. 139</i>)	10.7.4 Activity (10-13 min.)	Work in pairs to roll dice, record the data, and compare the results (<i>SE p. 142</i>)
	10.7.2 Guided Practice (10 min.)	Summarize data presented in different ways (<i>SE p. 140</i>)	10.7.5 Your Turn! (10-12 min.)	Compare and analyze two different data sets (<i>SE p. 143</i>)
	10.7.3 Try It (5-10 min.)	Compare data presented in different ways (<i>SE p. 141</i>)	10.7.6 Wrap-Up (~5 min.)	Explain how to compare a data set to a graphical representation of data (<i>SE p. 145</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Dice (one for each pair of students) - If dice are not available, have students use random.org and select one die as the option	(Optional) Consider choosing student pairs ahead of time based on data from prior lessons

 Teacher Tips	Common Misconceptions - Students may struggle to compare data sets without first calculating the measures of center and variation. - Students may struggle to explain their answers in writing.	Things to Consider - Consider providing writing prompts or sentence starters for questions that require written explanations. - If time is an issue, then consider assigning all or part of the 10.7.5 Your Turn! for homework.
	Literacy and Language Routines Think-Pair-Share 10.7.2 Guided Practice provides an opportunity to engage the class in a Think-Pair-Share routine. This will give students time to contemplate why comparing data graphically may be easier, while also considering the drawbacks of box plots in terms of information that can be collected. As students share with a partner, circulate and encourage students to build on to or evaluate their partner’s response.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts Throughout this lesson, students work collaboratively and independently to analyze data from real-world contexts. This encourages students to conceptually understand measures of center and variation and how they can be applied to make sense of data in the real world. Clarifications for this standard include providing students with opportunities to create models and perform investigations, which students are given the opportunity to do through the 10.7.4 Activity.
	 Differentiate Instruction Throughout this lesson students are provided with multiple entry points to engage with the content. Visual displays of data are presented throughout to help students make sense of the data sets. Written explanations are intentionally scaffolded through the use of word banks, and this will be particularly helpful for ELL students to reinforce the meaning of the terms through the connection to visual data displays. Partner work is built in throughout this lesson, providing auditory entry points and means for verbal processing of information through discussion.	
Lesson Notes:		

10.7.1 Warm-Up: Would You Rather?

Component Overview: Two deals are presented, and students explain which deal they would rather take.



10.7.1 Warm-Up: Would You Rather?

1. Would you rather...

- A. Pay for 3 gifts and get 1 of equal or lesser value free?
- B. Pay for 4 gifts using a coupon for 20% off?

Explain your reasoning.

Answers will vary. If the gifts cost \$5 each, a person selecting option A would pay \$15 and a person selecting option B would pay \$16. Option A would be the better choice.



Challenge students who finish early to prove why their choice is the right choice, using examples, and consider whether there are circumstances that would change which deal they would take.

10.7.2 Guided Practice: Summarizing Data

Component Overview: Students compare travel times for two counties given data sets represented in different ways.

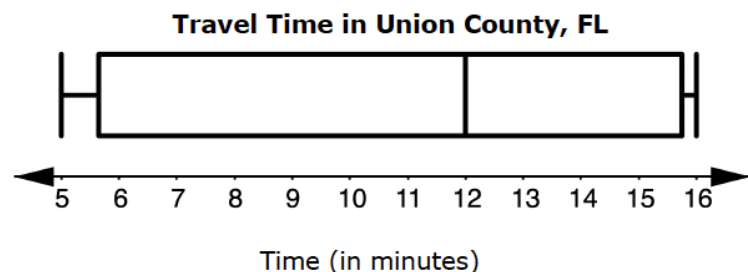


10.7.2 Guided Practice: Summarizing Data

- The commute times, in minutes, for 24 randomly selected people in Duval County, FL is shown.

9 10 15 17 18 20 20 21 22 23 24 25
25 25 25 26 26 29 29 29 32 34 38 40

The box plot displays the commute time, in minutes, for 24 randomly selected people in Union County, FL.



Think-Pair-Share

Pair students and engage the class in a Think-Pair-Share routine as they answer questions 1a and 1b. This will give students time to contemplate why comparing data graphically may be easier, while also considering the drawbacks of box plots in terms of information that can be collected. As students share with a partner, circulate and encourage students to build on to or evaluate their partner's response.

- Explain why it would be beneficial to create a box plot of the Duval County data when comparing these data sets.

Answers will vary. Having a box plot of Duval County would make comparing the data easier because both data sets will be in the same form.

- What are some measures used to describe data that cannot be determined from a box plot?

Answers will vary. Mean and mode

10.7.2 Guided Practice: Summarizing Data (continued)

- c. Use at least three terms from the term bank below to write a three-sentence summary comparing travel times to work in Duval County to travel times to work in Union County.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

Answers will vary. The median in Duval County is 25 minutes where the median in Union County is 12 minutes. It takes longer to travel to work in Duval County than it does in Union County. The IQR for Duval County is $29 - 20 = 9$ minutes, and the IQR for Union County is about 10 minutes, based on the box plot. The variability of the data can be compared through the range of each data set. The travel times vary less in Union County because the range is 11 minutes whereas the range in Duval County is 31 minutes.



Question 1c is scaffolded through the use of a word bank to provide students with key terms to consider as they explain their response. If you notice students struggling with their answer, further scaffold with questions like:

- Which measures of center can we find for each data set? What will this tell us?
- Which measures of variability can we find for each data set? What can this tell us?
- What can you say for sure about the travel times of each county? Why?

10.7.3 Try It: Comparing Data Sets and Graphs

Component Overview: Students work independently or with a partner to compare two data sets presented in different formats.



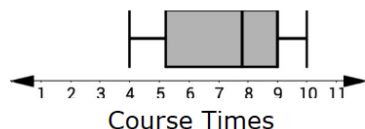
10.7.3 Try It: Comparing Data Sets and Graphs

- The Deltona High School Wolves and Wildwood High School Wildcats cross-country teams ran an obstacle course. Each team's times are summarized.

Deltona HS Wolves' Obstacle Course Times

4:25	4:43	4:49	5:02	5:12
5:21	5:31	5:32	5:37	5:52
5:54	6:08	6:20	6:26	6:33
6:48	6:53	7:16	7:23	8:05

Wildwood HS Wildcats' Obstacle Course Times



- Which statements are true about the data for the Deltona Wolves and the Wildwood Wildcats? Select all that apply.
 - ☐ The median time of the Deltona Wolves is less than the median time of the Wildwood Wildcats.
 - ☐ The fastest 25% of athletes on both teams complete the obstacle course in about the same amount of time.
 - ☐ The interquartile range of the Deltona Wolves is less than the interquartile range of the Wildwood Wildcats.
 - ☐ Approximately 50% of Wildwood Wildcats have times between 5 and 6 minutes.
- Which team do you think is more likely to win the next race? Explain your reasoning.

Answers will vary. Based on the data, Deltona will win the next race. Their course times vary less than Wildwood and all fall under Wildwood's 10 minute mark. Analyzing each team's IQR, Deltona's spread is smaller.



Students may struggle to compare the data sets, particularly when reasoning about the measures of center and variation to choose the true statements. Use strategic questioning to scaffold this section for students:

- How can you determine the median of the data sets?
- Where are the fastest 25% of athletes located in each data set?
- How can you determine the IQR for both teams?



It may be helpful for students to visualize Deltona's times as "quartiles," similarly to how data is presented in the box plot. The data is organized into 4 rows increasing in times from left to right. Encourage students to circle or label each row. The first row corresponds to the left whisker of a box plot, the second row corresponds to the first quartile, the third row corresponds to the third quartile, etc. Though this is not exact, the visual and labels can be a starting point for students as they reason about which statements are true in question 1a.

10.7.4 Activity: Roll the Dice

Component Overview: Students work in pairs to play “Roll the Dice.” Each partner rolls the die 25 times, and both partners record the results either on a line plot or numerical data set before comparing their data sets.



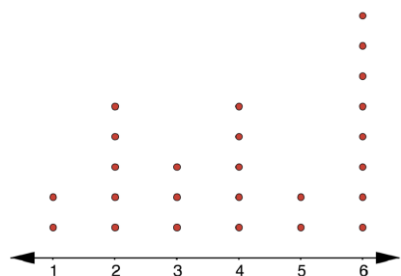
10.7.4 Activity: Roll the Dice

With your partner, decide which of you will be “Partner 1” and “Partner 2.” Partner 1 will be creating a line plot and Partner 2 will be recording a data set.

Using a standard six-sided die, each person will roll one die 25 times. After each roll, record the results in either a line plot or numerical data set below, depending on if you are Partner 1 or Partner 2.

- When both partners are finished, copy their data below.
Answers will vary.

Line plot (Partner 1’s rolls): *Answers will vary.*



Data set (Partner 2’s rolls): *Answers will vary.*

1	2	6	3	6
6	2	4	6	6
6	1	3	5	6
3	5	5	5	5
1	2	5	4	4



If dice are not available, have students use random.org and select one die as the option.



This activity provides auditory, visual, and kinesthetic entry points. Learners have a variety of ways to engage with the content and expand their understanding of how to compare data sets.

10.7.4 Activity: Roll the Dice (continued)

2. Complete the table by calculating the measures of center and spread for your data. Share your data with your partner and add their data to the table.

Answers will vary.

	Partner 1	Partner 2
Mean	3.96	4.08
Median	4	5
Range	5	5
IQR	4	3.5

3. With your partner, discuss similarities and differences in your data. Summarize your discussion in the table.

Answers will vary.

Similarities	Differences
<i>The range for both is 5. Both partners rolled 25 times. The means are both close to 4.</i>	<i>Partner 2 had a larger median but a smaller IQR.</i>



Consider providing guiding questions for students to reference as they discuss the similarities and differences between their data sets. This will encourage participation and help guide thinking towards deeper understanding.

- How do the means of the data sets compare? What does that tell you?
- What are the medians of each data set? What does that tell you?
- Which measures of center/variation best summarizes each data set? Why?

10.7.5 Your Turn!

Component Overview: Students work independently to compare data sets represented in different ways.



10.7.5 Your Turn!

1. A group of ten rowers that practice outdoors on the water (Group 1) are challenging a group of ten who use indoor rowing machines (Group 2). They want to see who can row faster using rowing machines.

The time, in minutes, it took each of the ten rowers in Group 1 to row 500 meters on the rowing machine is represented in the stem-and-leaf plot.

Group 1: Outdoor Rower Times (in minutes)

Stem	Leaf	
15	2, 8	<i>Mean: 1.733 minutes</i>
16	8	<i>Median: 1.765 minutes</i>
17	0, 5, 8	<i>Range: 0.35 minutes</i>
18	0, 2, 3, 7	<i>IQR: 0.14 minutes</i>

Key: 15|2 = 1.52



After students have read the context, consider prompting them to determine the measures of center and variation for each data set to prepare them for the questions that follow.

The time, in minutes, it took each of the ten rowers in Group 2 to row 500 meters on the rowing machine is represented in the data set.

{1.63, 1.82, 1.95, 1.97, 2.0, 2.02, 2.12, 2.20, 2.25, 2.88}

Mean: 2.084 minutes
Median: 2.01 minutes
Range: 1.25 minutes
IQR: 0.25 minutes

10.7.5 Your Turn! (continued)

For each statement below, state whether it is true or false. For any statements that are true, explain why.

Statement	True or False?	Explanation
The center of the Group 1 times is similar to the center of the Group 2 times.	<i>False</i>	<i>The center for Group 1 is lower than Group 2. Both the mean and median for group 1 is smaller than the mean and median, respectively, for Group 2.</i>
The variability of the Group 1 times is similar to the variability of the Group 2 times.	<i>False</i>	<i>The variability is different for the groups, regardless if the range or IQR is used to compare. Both the range and IQR are smaller for Group 1 compared to the same measure for Group 2.</i>
The Group 1 rowers that practice on the water are typically faster than the Group 2 rowers that row on machines.	<i>True</i>	<i>The rowers that practice on water, Group 1, have faster times than those who row indoors, Group 2. The measures of center for Group 1 are smaller than Group 2, indicating a faster time.</i>
The Group 1 rowers that practice on the water are more consistent than the Group 2 rowers that row on machines.	<i>True</i>	<i>Answers will vary. Both measures of variability for Group 1 are low, indicating their times are not as spread out from one another.</i>



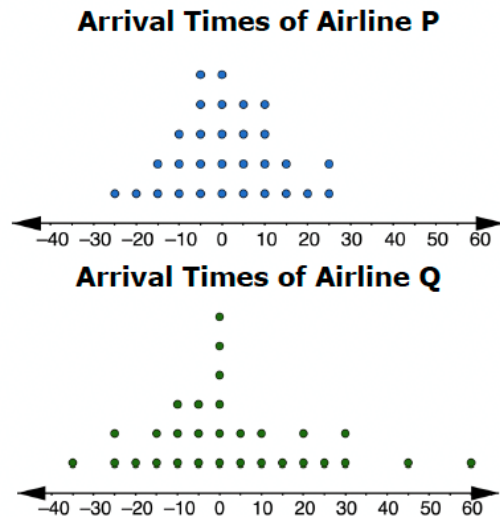
Students do not need to write explanations for the false statements. Sample responses are provided for reference, though.



The students are not directly told to calculate center and variability. Instead, they are asked questions about the summary of the data sets. If students are struggling, it may be helpful to have them determine the measures of center and variability for each data set.

10.7.5 Your Turn! (continued)

2. The dot plots display the arrival times of 30 flights for two different airlines.



- A negative number represents the number of minutes the flight arrived before its scheduled time.
- A positive number represents the number of minutes the flight arrived after its scheduled time.
- A zero indicates that the flight arrived at its scheduled time.

Based on the data, from which airline would you choose to buy your ticket? Use measure(s) of center and variability to justify your choice.

Answers will vary.

Airline P – mean: 1, median: 0, range: 50, IQR: 15

Airline Q – mean: 3.17, median: 0, range: 95, IQR: 25

Airline P has less variation with an IQR of 15 compared to Airline Q which has an IQR of 25. Airline P also has an average arrival time of 1 minute after their scheduled time where Airline Q has an average arrival time over 3 minutes after their scheduled arrival time. Based on the data, Airline P would be the better choice to buy a ticket from.



Students may struggle to understand what negative numbers and 0 mean in the context of the flight times and therefore choose the airline with the higher measures of center. If you notice students making this mistake, ask questions to foster contextual understanding:

- *Have you ever flown on a plane? Did you arrive on time?*
- *Is it better to have a flight time longer or shorter than expected? Why? Which numbers indicate longer flight times? Shorter?*
- *If you want to arrive on time or before schedule, which airline should you choose based on the data? Why?*

10.7.6 Wrap-Up: Cool Down

Component Overview: Students summarize their experience from the lesson.



10.7.6 Wrap-Up: Cool Down

1. Write a short note to a classmate who missed class summarizing today's lesson.

Answers will vary.



Consider pairing students to discuss their notes. Use student responses as anchor charts or added to scaffolding notes.